

Algebra Commutative Practice Quiz 3b

$$f(x) = \sin^2(x)$$

i	x_i	$f(x_i)$	$f'(x_i)$	$f(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$	$f(x_1, x_2, x_3, \dots, x_n)$	$f(x_1, \dots, x_{i-1}, x_{i+1}, \dots, x_n)$
0	0	0	$2/\pi$	0	$-32/3\pi^3$	$64/3\pi^4$
1	$\pi/4$	$1/2$	$2/\pi$	$-8/\pi^2$	$32/3\pi^3$	/
2	$\pi/2$	1	$-2/\pi$	0	/	/
3	$3\pi/4$	$1/2$	$-2/\pi$	/	/	/
4	π	0	/	/	/	/

$$\frac{1/2 - 0}{\pi/4 - 0} = \frac{1/2}{2\pi} = \frac{1}{4\pi} \quad \frac{1/2}{\pi/4} = \frac{2}{\pi} \quad \frac{-1/2}{\pi/4} = -\frac{2}{\pi} \quad \frac{-1/2}{\pi/4} = -\frac{2}{\pi} \quad \frac{64}{3\pi^3} = \frac{64}{3\pi^4}$$

$$\frac{-1/4}{3\pi/4 - \pi/4} = \frac{-1/4}{\pi/2} = -\frac{1}{2\pi} \quad \frac{-1/4}{\pi/2} = -\frac{1}{2\pi} \quad \frac{-1/4}{\pi/2} = -\frac{1}{2\pi} \quad \frac{-1/4}{\pi/2} = -\frac{1}{2\pi} \quad \frac{-1/4}{\pi/2} = -\frac{1}{2\pi}$$

$$\text{so } p_4(x) = 0 + \frac{2}{\pi}(x-0) + 0(x-0)(x-\frac{\pi}{4}) - \frac{32}{3\pi^3}(x-0)(x-\frac{\pi}{4})(x-\frac{\pi}{2}) + \frac{64}{3\pi^4}(x-0)(x-\frac{\pi}{4})(x-\frac{\pi}{2})(x-\frac{3\pi}{4})$$

$$x^2 - \frac{\pi}{4}x$$

assembly!

$$\rightarrow x^3 - \frac{3\pi}{4}x^2 + \frac{\pi^2}{8}x$$

$$x^3 - \frac{3\pi}{4}x^2 + \frac{\pi^2}{8}x$$

assembly!

$$\rightarrow x^4 - \frac{3\pi}{2}x^3 + \frac{11\pi^2}{16}x^2 - \frac{3\pi^3}{32}x$$

$$= \frac{2}{\pi}x - \frac{32}{3\pi^3}(x^3 - \frac{3\pi}{4}x^2 + \frac{\pi^2}{8}x) + \frac{64}{3\pi^4}(x^4 - \frac{3\pi}{2}x^3 + \frac{11\pi^2}{16}x^2 - \frac{3\pi^3}{32}x)$$

$$= \frac{2}{\pi}x - \frac{32x^3}{3\pi^3} + \frac{8x^2}{\pi^2} - \frac{4x}{3\pi} + \frac{64x^4}{3\pi^4} - \frac{32x^3}{\pi^3} + \frac{44x^2}{3\pi^2} - \frac{2x}{\pi}$$

$$= \frac{64x^4}{3\pi^4} - \frac{128x^3}{3\pi^3} + \frac{68x^2}{3\pi^2} - \frac{4x}{3\pi}$$